

Processes for Developing Customer-Centered Sites

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This chapter takes the patterns, principles, and techniques of customer-centered design as described in Chapters 2 through 4 and places them in the context of a complete Web site design process. Think of this chapter as a rough guide to designing, implementing, and maintaining a Web site. You probably have a design process that you use today, and you might say that it's good enough. What we offer is not a quick fix, but a program that will make any Web site you design more useful, usable, reliable, and satisfying for customers.¹

Our goal is to provide a general process that you can use when creating or updating a Web site—something that will help you focus your time and energy on clear goals. A well-defined process is also useful for your clients. It lets them know what they can expect from you and what you need from them to build a Web site that meets their expectations and the needs of their customers.

The design process will not always go as smoothly as described here. It is *iterative*; that is, it repeats and jumps back and forth when necessary. Nor will this process solve all your problems. Tailor it to your team, your project, and your organization. Formal procedures that are necessary for large teams may be overkill for small teams. Techniques that work for art-centered design firms are unlikely to work for e-commerce-centered design firms. At a minimum, however, include in your process the major activities discussed in this chapter—that is, the first four phases of development, as defined next.

¹ Again, we use the term *customer* to mean any person who will use the Web site that you're designing, whether it's a business or a government site, and whether the person is an end customer, employee, site administrator, or partner. We use the term *clients* to refer to the people for whom you're doing the work, the people providing the funding.

Development Process Overview

5.1

Generally speaking, development of a Web site can be broken down into seven phases (see Figure 5.1):

- 1. Discovery.** Understanding the target customers and their needs, and conceptualizing the business and customer goals for the Web site.
- 2. Exploration.** Generating several rough initial Web site designs, of which one or more will be chosen for further development.
- 3. Refinement.** Polishing the navigation, layout, and flow of the selected design.
- 4. Production.** Developing a fully interactive prototype and a design specification.²
- 5. Implementation.** Developing the code, content, and images for the Web site.
- 6. Launch.** Deploying the Web site for actual use.
- 7. Maintenance.** Supporting the existing site, gathering and analyzing metrics of success, and preparing for the next redesign.

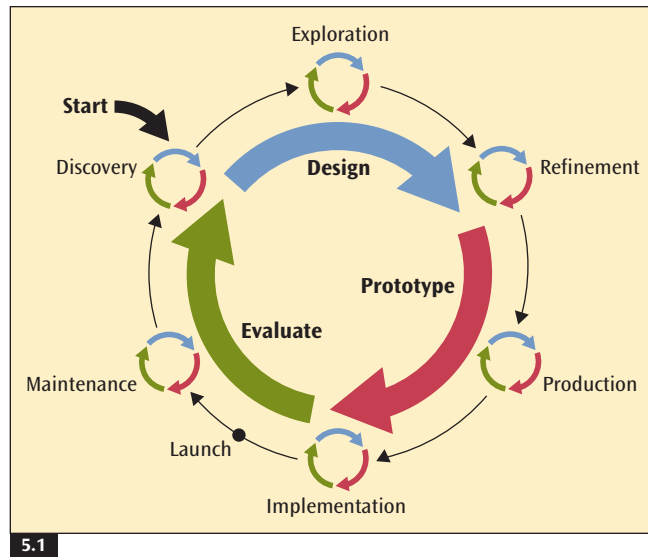
The first four steps, Discovery through Production, focus on the overall design of a Web site, clarifying what customers can do on the site and how they do it. You might describe these four steps as *the* design process. Each is characterized by rapid iteration with progressive refinement, moving the design from high-level and general to increasingly specific and detailed. During these stages we have found that the more time you spend up front in the tight iterations, the more likely it is that the Web site will meet customer expectations. In the Discovery phase a team might iterate five to ten times or more on paper. As the team moves into the electronic representations used in the Refinement stage, it might iterate much less, perhaps only three or four times. The exact number of iterations depends on how well the design performs when evaluated.

Punctuate each of the first four steps with a presentation to the client. Hand over any other agreed-upon deliverables at this time, such as a site map, a high-level diagram of a Web site, or a specification document detailing what the Web site will do when it is completed. The main point

² Note that *production* in this case does not mean creating the site but refers instead to creating the “blueprint” for the site so that someone can build it. Some design firms use the term *production* to mean the actual creation of the Web site—that is, what we have termed *implementation*. However, most designers from our interviews used *production* in the same sense that we have defined it here, so we have kept the term.

Figure 5.1

The Web site development process, both as a whole and at each individual phase, uses iterative design.



5.1

of the presentation, though, is to obtain approval about the work you performed during that phase.

These presentations should be only one part of what will now have become an ongoing dialogue with the client. Several parties have a stake in the design process, including the client and the intended customers. At many points in the process the design team can and should report back to the client to ensure that it's on the right track. For many design consultancies, the client calls the shots, so the client's needs and requirements are considered on a regular basis.

The last three phases—Implementation, Launch, and Maintenance—may not be part of your development process. In fact, many design firms stop at Production, handing off all documentation and interactive prototypes to another team, often in the client's organization but sometimes in a third firm, which goes on to do the rest of the work.

We do not include detailed planning at this stage for two reasons: First, each organization has its own way of handling issues such as scheduling, budgets, and risk management. Second, *the process is by no means linear*. A team may be in the Exploration stage and discover that it needs to do more Discovery activities. Another team may be in Production and realize that it needs to go back to Refinement or Exploration to hammer out some details. Such is the nature of design. Although going backward may seem like a defeat of sorts, take heart in the knowledge that time and money are actually being saved if changes are made sooner rather than later.

The Discovery Phase

5.2

The purpose of Discovery is to determine and clarify the scope of the project, the business goals of the client, and the needs of the intended customers. This phase often starts with analysis of whether a Web site is the best solution for the client's goals. By the end of Discovery, you and your client will have a shared understanding of three things:

1. The target customers and their needs (people, tasks, technology, and social issues), which are described in the **customer analysis document**
2. The business goals of the project, described in the **business analysis document**
3. The features that the Web site should provide for customers when the work is completed, described in the **specification document**

You will deliver these three documents to the client by the end of this phase. The process of creating these three documents is also known as **requirements gathering**. The sections that follow look in more detail at each of the steps of Discovery and the documents that result from this process.

The Discovery Process

Step 1: Determine the Overall Goals of the Web Site • You and your client will work together to answer questions before creating a new Web site or redesigning an existing site. These questions might include the following: What value will the Web site provide to customers? What will the site accomplish? Is the goal of the site to sell products online, promote products, educate, inform, provoke, communicate, or provide a community? What will be the focus of the site? *Why should the site be built at all?*

Other crucial questions that must be asked are these: What value will the Web site provide to the client? What role does the site play in relation to the rest of the company? How will the site further the client's overall goals? The design team often starts this phase by defining the site together with the client, focusing on the client's goals, the high-level services the client wishes to offer, and preliminary customer categories. These decisions form the basis for the rest of the Discovery process.

Some techniques that might be applied at this phase include interviewing or corresponding with clients to clarify what they expect. Discover the needs of the site's target customers by interviewing, running focus groups

- 3 with, and surveying target customers online and offline. These techniques are described in greater detail in Chapter 3 (Knowing Your Customers: Principles and Techniques).

This is also a good time to evaluate the existing Web site, as well as to review and evaluate competitors' Web sites for opportunities to improve and differentiate your site from theirs. Techniques for these evaluations are described in Chapter 4 (Involving Customers with Iterative Design).

4

- Step 2: Decide on the Web Site's Value Proposition** • One of the most important patterns that come into play here is the UP-FRONT VALUE PROPOSITION (C2). The value proposition states what the Web site offers to the target customers. The goal is to explain the Web site's value convincingly to someone on the street, in one sentence. Everything you will put on the Web site draws on this single idea; it is the theme that unifies the site. Figure 5.2 shows compelling value propositions from some actual Web sites.

If you're creating a new Web site, think through the value proposition carefully, making many drafts before you decide on the best one. If you're updating or redesigning an existing Web site, you may need to reconsider the existing value proposition. In either case the UP-FRONT VALUE PROPOSITION (C2) pattern provides steps for hammering out the right one.

C2

You will also consider initial branding ideas at this early design phase. What do you want people to think of when they think of your Web site? What kinds of feelings do you want people to have after visiting your site? Do you want to be thought of as reliable and trustworthy, or maybe

Figure 5.2

Many Web sites make their value propositions clear from their taglines. The goal of the value proposition is to provide a single, powerful idea of what the Web site is all about.



5.2



(www.shutterfly.com, www.slashdot.org, www.firstgov.gov, 2001)

exciting and fun? The **SITE BRANDING (E1)** pattern provides exercises to help you decide what kind of impressions to give customers.

E1

At this stage, start thinking about whether personalization is useful for individual customers. The **PERSONALIZED CONTENT (D4)** pattern provides some starting points. For example, if you're working on an e-commerce site, consider how personalization can help your customers find interesting and useful products. Maybe your Web site can recommend products, or maybe it can let people see a list of the most popular items. If you're developing a news site, perhaps your visitors can specify zip codes to get local news and weather.

D4

Speaking of zip codes, this is a good time to bring up the issue of internationalization, because some countries outside of the United States do not use zip codes. Again, it comes back to the question of defining your customers. Do you expect people from other countries to use your Web site? If so, there are a host of issues to consider, including currency, color, icons, and layout. We discuss many of these issues in more detail in the **INTERNATIONALIZED AND LOCALIZED CONTENT (D10)** pattern.

D10

Be Firm about Not Skipping Discovery • Some clients may insist on skipping the Discovery phase and jumping straight to Web site development. Unless the scope of the project and the needs of the customers have already been defined, this is usually a bad idea and a likely prescription for failure. It is your job to explain the importance of this phase. [We can help. See Chapter 3 (Knowing Your Customers: Principles and Techniques) for some convincing arguments.]

3

Avoid Gold-Plating the Web Site • The term **gold-plating** means trying to get the Web site absolutely perfect before deploying it. You have probably visited other Web sites, seen something cool, and have kept saying to yourself, "We should have this on our Web site too!" Usually that cool thing is technically complex (and fun) to design and implement but, frankly, may not be all that helpful to customers.

You can avoid the problem of investing a lot in something that has little gain for the customer by getting continuous feedback from customers about what's useful and what's not, and prioritizing the features on the basis of that feedback. This strategy will help you cut through the morass of features and keep you focused on what needs to be done. You do not have to develop and deploy all the features simultaneously. Deploy the features in stages, in many small steps instead of one big leap. Plan for future growth, but also plan for the next deployment.

G3
G4
F3 F1

Get the Web Site Fundamentals Right First • The last thing to watch out for is not taking care of the fundamentals first. This becomes an issue when you design things out of order. For example, for e-commerce sites it does not make sense to design and implement **PERSONALIZED RECOMMENDATIONS** (G3) or a **RECOMMENDATION COMMUNITY** (G4) if there are still problems with the **SHOPPING CART** (F3) and the **QUICK-FLOW CHECKOUT** (F1). Take care of the things that the Web site absolutely must have before adding the icing that makes it more attractive.

Discovery Deliverables

Design teams produce three main documents as an outcome of the Discovery phase: the customer analysis document, the business analysis document, and the specification document. Let's look at each one in turn.

Customer Analysis Document • This document gives the design team and the client a deep understanding of and empathy for the Web site's intended customers. It fleshes out the intended customers, describing their characteristics, their needs, and their tasks. It includes the following details:

- The motivation of customers to visit the Web site, or the **UP-FRONT VALUE PROPOSITION** (C2).
- A **task analysis** of the intended customers, describing the people, their tasks, the technologies they use, and their social and organizational issues. One increasingly popular way to do this is by creating personas, or highly detailed fictional people, who are representative of the customers. Giving the customers names makes it easier to talk about them.

3 We provide more details about customer analysis in Section 3.2, Techniques for Knowing Your Customers, in Chapter 3 (Knowing Your Customers: Principles and Techniques).

Business Analysis Document • This document spells out the business needs of the client and the business goals of the Web site. It explores how the goals of the client map to the tasks and customers discovered during the task analysis. For example, suppose the client's goal for an intranet site is to be the primary source of company information. What does this goal mean to an administrative assistant who is just trying to find the information he needs to get his job done?

If you're revising an existing Web site, a new business analysis is probably not necessary, but it is still a good idea to check every so often that

the business goals are the right goals, and that the Web site works toward those goals. A business analysis document usually includes the following:

- **Business plan.** This plan describes the business goals of the Web site and the client's needs. Some goals might be to support existing customers, to bring in new customers by providing information about products, and to increase sales by enabling purchases online.
- **Competitive analysis.** This analysis determines the features that competitors have on their Web sites, and it identifies which features are important to customers and which are not. It also discusses the competitive advantages that the proposed Web site will have over others, and it expresses these advantages as high-level goals.
- **Metrics for success.** How will success be measured for both the business and the competitive goals? For example, how many customers does the site need to draw to stay in business? But just as important as attracting customers is retaining them, keeping them coming back for more. How many are repeat customers? What is the conversion rate; that is, how many visitors become paying customers? How many become community members?

Specification Document • Also known as a *requirements document*, the specification document identifies what the Web site should provide when the work is complete. It describes any functionality that the Web site needs, as well as any constraints on the system. At this point you do not have to start thinking about how you'll achieve the needed functionality. Focus instead on what you'll accomplish. A specification document contains the following:

- **Project description**, describing the common purpose and ultimate goals of the project, from both client and customer perspectives.
- **List of tasks, scenarios, and storyboards**, fleshing out the features [see Chapters 3 (Knowing Your Customers: Principles and Techniques) and 4 (Involving Customers with Iterative Design) for more information]. These tasks will form the basis of the Web site evaluations. The number of tasks depends on the complexity of the proposed work. Simple projects can make do with ten to twenty complete tasks, but larger projects will need enough to cover all the proposed features. Label tasks as *easy*, *moderate*, or *difficult*. Customers should be able to complete all of the easy and moderate tasks and many of the difficult ones.
- **Comprehensive list of proposed features**, classified in importance as "must have," "should have," or "could have." Use competitive comparisons, as well as surveys and other market research techniques, to

obtain this type of information. Decompose features into subfeatures. For example, a Web site that helps manage personal information will likely have a contact manager, which lets people add new contacts, edit existing ones, and search for contacts by name. Each feature also includes a short statement on how it will be evaluated or tested in the final Web site.

- **Overall design goals**, such as reducing the number of mistakes that customers make on the existing site, decreasing the time it takes to make purchases and check out from the shopping cart, or making the site faster to use.
- **Metrics** to measure whether the team has reached these goals and requirements, such as keeping download time to less than five seconds for 90 percent of the target customers. State how each feature will be measured in the final Web site. (A more precise test specification will be developed later, spelling out more of the details.)

Not writing a specification document is the biggest risk you can take, yet this is the most often skipped step. A specification document does not have to be long or formal, but it is useful because it forces you to think through important details and make sure that they make sense and are realistic. A specification document also makes it easier to communicate with clients and with other team members so that everyone has a shared vision of what the Web site will be like when completed.

Specification documents are often tedious to read. Be brief and concise, and use lots of diagrams to illustrate what you mean. Make it interesting enough that you would take the time to read it yourself.

5.3 The Exploration Phase

During the Exploration phase you will generate and explore several designs. These initial designs often do not reflect ideas about color, imagery, and typography. However, they do reflect ideas about site structure and navigation. By the end of Exploration, you will have several prototypes to present to the client, who will select one for further development and sign off on the work done. Sometimes a client will want to fund continued development of two sites for further refinement before making a final choice. In either case, the selected design is supported by evaluation results that show it is the best at meeting the business, client, and customer goals.

The Exploration Process

Typically you will generate medium-fidelity site maps, storyboards, and schematics [see Chapter 4 (Involving Customers with Iterative Design)

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for details about medium-fidelity prototypes, and about the differences among site maps, storyboards, and schematics]. Test all your designs quickly with target customers to ensure usefulness and usability.

More design patterns come into play here too. The **HOMEPAGE PORTAL (C1)** pattern describes some of the ways to structure your homepage, as well as what you'll want on your homepage, such as a **PRIVACY POLICY (E4)** and a **SEARCH ACTION MODULE (J1)**.

Begin initial work on the information architecture—the overall organization of the Web site's content—at this phase as well. The **BROWSABLE CONTENT (B2)** pattern provides more details on how to design and implement your architecture.

C1

E4

J1

B2

Exploration Deliverables

Medium-Fidelity Site Maps, Storyboards, and Schematics • At the end of the Exploration phase you will present several sets of medium-fidelity site maps, storyboards, and schematics to your client. Each set represents a design alternative that addresses the issues described by the customer analysis document, the business analysis document, and the specification document. In particular, the storyboards will show the initial ideas for how the scenarios in the specification document will be carried out. However, at this phase none of the deliverables will have much detail; rather they will have just enough detail to represent the general idea. Figure 5.3 shows an example of a medium-fidelity site map.

The Refinement Phase

After you have chosen a design idea from the variations presented in the Exploration phase, you need to develop the selected idea further. Polish the navigation, layout, and flow of the selected design more, providing a clearer understanding of how the Web site will look and feel. By the end of Refinement, you will have a highly detailed prototype to present to your client, and you'll probably expect the client to sign off on the work.

The Refinement Process

During the Refinement phase, iteratively refine, detail, and informally test the design. Determine aspects such as the precise typeface of labels and body text, the exact sizes and appearances of images, and color schemes and palettes. For most sites you will not find it necessary to design every page at this stage, because you will break down the site into

5.4

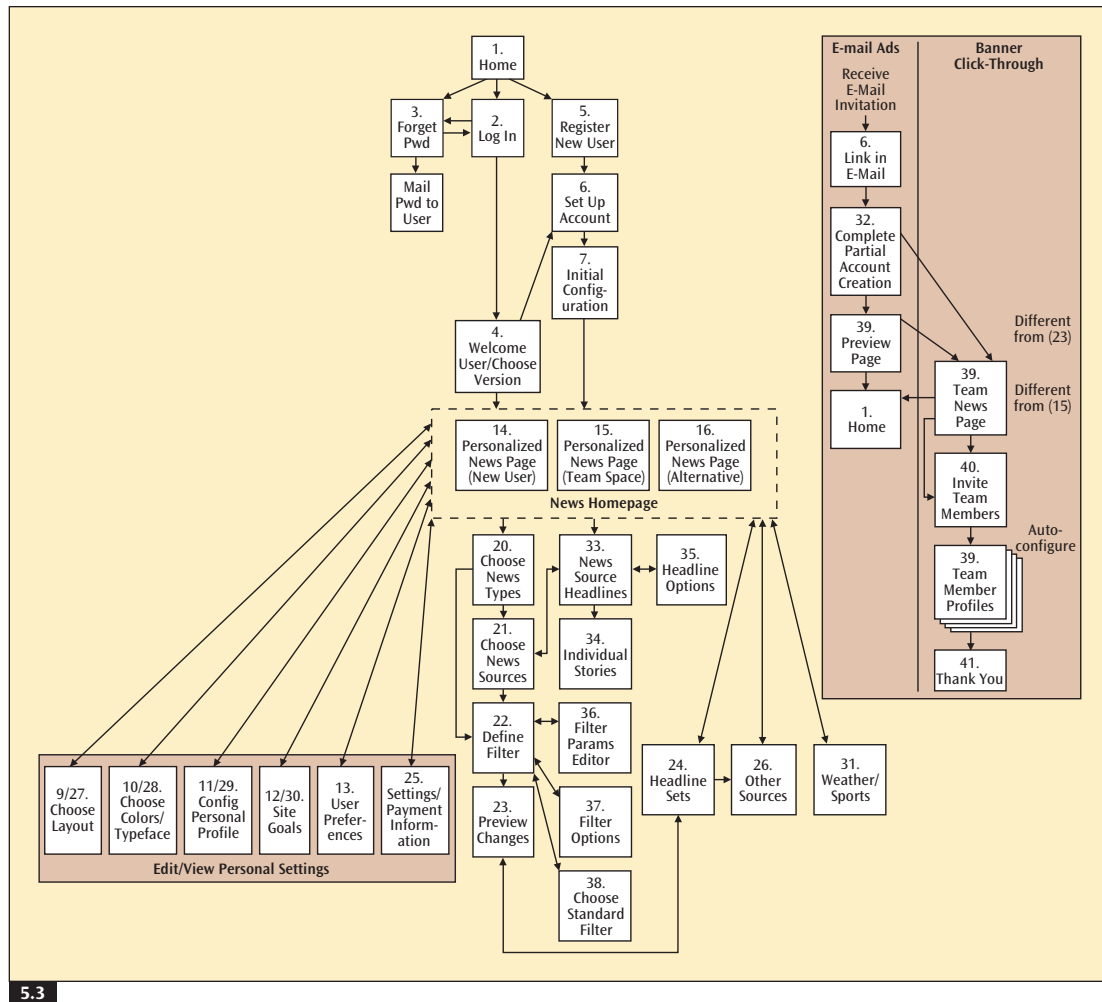


Figure 5.3

Medium-fidelity site maps such as this one can quickly communicate the flow of a site in an easy-to-read format.

classes of pages (such as homepage, second-level pages, and pages for specific types of content). An example, or **template**, can represent each of these classes of pages.

In this phase, site maps are still abstract representations of the entire Web site. On the other hand, storyboards and schematics are no longer drab and dull; instead they are now rich with images, icons, typography, and sophisticated color schemes.

The main difference between Refinement and Exploration is that the design you create in the Refinement phase has more detail than the designs you generated during the Exploration phase. You can apply many

of the same patterns in both phases. Some patterns are more useful during Refinement than during Exploration. For example, the **CLEAR FIRST READS (I3)** pattern takes a concept from graphic design, one intended to give a quick first impression of a visual design. It suggests that you explicitly design the first thing that a potential customer sees on a Web page to give an overall feel for the content of the page. A related pattern is **GRID LAYOUT (I1)**, which gives you ways of structuring Web pages in a consistent and understandable manner.

I3

I1

Another pattern to consider is **MULTIPLE WAYS TO NAVIGATE (B1)**. People navigate Web sites in many ways, using a variety of search mechanisms, text links, buttons, and navigation bars. This pattern shows how to provide multiple and sometimes redundant ways of navigating, to make it easier to find the right page.

B1

Refinement Deliverables

Medium- to High-Fidelity Site Maps, Storyboards, and Schematics • At the end of the Refinement phase, you will present one set of medium- to high-fidelity site maps, storyboards, and schematics to your client. These deliverables are similar to the ones in the Exploration phase, but they have much greater detail. For example, the site maps flesh out the overall site structure in detail, and the storyboards and schematics make use of more graphical images and color.

The Production Phase

5.5

The goal of Production is to create a detailed set of deliverables that embody and represent the final design idea. The deliverables, including interactive prototypes, written design descriptions, design guidelines, and technical specifications, are high-fidelity and contain as much detail as possible about the layout, navigation, visuals, and content for each Web page. Exactly what is delivered at the end of this phase depends on whether you will continue to the next phase (Implementation) or hand off the design to someone else.

If you will be handing off the design, make the interactive prototypes and specifications precise and highly detailed so that there's no ambiguity about what the next team will implement. You may want to call for more evaluation during Implementation to ensure that the specification is being implemented correctly. The client should review the ongoing development to catch any problems before the new team does too much work.

Ideally, though, the design team will work hand in hand with the engineering, art, marketing, editorial, and management teams so that the site is implemented as designed and tested, and so that the inevitable questions that arise can be answered.

The Production Process

During the Production phase, continue evaluating the product with real customers. Because this will be the first time that the entire design comes together, often with more complexity than existed in earlier prototypes, new interaction issues may arise.

During Production the patterns become more low-level as the problems become increasingly technical. For example, one pattern that will be used here is ACTION BUTTONS (K4), which explains how to make buttons that look like they can be pressed. Another useful pattern at this stage is OBVIOUS LINKS (K10), which discusses why links must be easy to see and explains how to make them that way.

Production Deliverables

The deliverables of the Production phase vary. Here we list the most common ways of describing the design in detail.

Design Document • In contrast to the specification document, the **design document** describes in great detail how the Web site works. It takes all of the features from the specification document and uses site maps, storyboards, and schematics to describe the flow of interaction.

If your work stops here and you hand the project off to another team, the design document needs to be detailed, descriptive, and unambiguous. If the same design team continues to do Implementation, the design document does not need to be so detailed. Provide just enough information that the client and the team understand precisely how customers will interact with the Web site, and how and when the site is complete.

Interactive Prototypes • Often delivered along with the design document, an **interactive prototype** gives you and your client a better grounding for how the finished Web site will look and behave. Create interactive prototypes using standard Web site production tools, such as Adobe GoLive or Dreamweaver, as well as prototyping tools such as Macromedia Director. The idea is to provide enough detail that everyone can see how the final Web site will look and feel. (For example, not all of the links have to work correctly, as long as it's clear what they're intended to do.)

Some firms deliver interactive prototypes in lieu of a design document but embed many of the details as annotation to the prototype. For example, moving the mouse over certain sections of the prototype might pop up a specification of the font family and size.

Technical Specifications • Technical details are explained in this document, which includes things like the kind of Web server, the kind of programming and scripting languages, the kind of database, and the version of HTML that will be used. Performance metrics can also be included, such as the number of customers who can be supported simultaneously. Optionally, the technical specifications might include some engineering prototypes as a proof-of-concept demonstrating that the difficult parts of the proposed design are technically feasible and can realistically be built.

Design Guidelines • These guidelines are the general rules to be followed on every Web page to minimize inconsistencies between pages. A design guideline could describe which fonts should be used, what color the links should be, when the logo should be used and where it should be positioned, what color scheme to use, what the maximum file size of each Web page is, and so on. The person who will implement and/or maintain the Web site should be able to understand and use the guidelines. Optionally, the guidelines might also include a style guide to ensure that the writing is consistent throughout the Web site, for example, using the term *e-mail* instead of *email*.

Web Page Templates • Web page templates are the HTML files that represent typical Web pages on the site. The goal of using these templates is to avoid inconsistencies between pages. With a minimum of effort you can copy and modify these templates with content specific for a particular page. Six patterns that are useful to consider when you're developing Web page templates are PAGE TEMPLATES (D1), GRID LAYOUT (I1), ABOVE THE FOLD (I2), CLEAR FIRST READS (I3), EXPANDING SCREEN WIDTH (I4), and SITE ACCESSIBILITY (B9).



The Implementation Phase

The aim of Implementation is to create the HTML, images, database tables, and software necessary for a polished and fully functional Web site that can be rolled out and used by its target customers. Making sure that you've reached this goal requires running formal usability tests to

5.6

ensure that customers can complete the tasks they want to do. The quality assurance group, both individually and as a whole, will test all the code, graphics, and HTML thoroughly so that the Web site works as intended and downloads quickly. [See the LOW NUMBER OF FILES (L1), FAST-LOADING IMAGES (L2), and FAST-LOADING CONTENT (L6) patterns in Pattern Group L (Speeding Up Your Site).] Check all the content for accuracy at this stage.

The Implementation Process

Devote more effort to content in the Implementation phase. Two patterns that come in handy here are WRITING FOR SEARCH ENGINES (D6) and DISTINCTIVE HTML TITLES (D9), which look at how to improve the internal structure of Web pages for search engines and for customers. Another useful pattern is INVERTED-PYRAMID WRITING STYLE (D7), which describes techniques for making text content easier to skim and faster to read. On sites with content that changes frequently, a useful pattern for building a content management system is CONTENT MODULES (D2), which put content in a database to keep production costs low and site reliability high.

Tools useful at this stage include a revision control system for storing and sharing files among a group of people, and a bug database for tracking problems. Choose tools that you already know how to use and that have been proven to work. The Implementation phase of development is a bad time to try out an unknown product. (However, do try out new tools between projects, to keep your skills sharp and up-to-date.)

You also need to determine naming conventions for folders and files. For example, which folder will contain the images? Will there be just one folder or many? If you're selling products, what will be the name of each product's CLEAN PRODUCT DETAILS (F2) page? Is the naming set up to make it easy to add new product pages in the future?

If you're using SECURE CONNECTIONS (E6), you may need to apply for digital certificates at this time. **Digital certificates** are a way for Web servers to prove that they really are who they say they are, and they are issued by a variety of trusted third-party vendors known as **certification authorities**. Keep in mind that digital certificates are often bound to a specific server name, so you may encounter problems if you try to move a digital certificate from a test server to the actual Web server. In this situation, it might not be a bad idea to get multiple certificates, for all the Web servers that you will be using.

The Web site also needs to be rigorously tested in this phase to ensure that it is high-quality and professional. This testing consists of performing

more usability tests, doing some editorial spot-checking, and running automated test suites. The following checklist identifies some of the tests that need to be done during Implementation:

- Check that the Web site has all of the features stipulated in the specification document, and that the features are implemented correctly.
- Check that the developed Web pages are compatible with various Web browsers, including text-only browsers used by the blind.
- Test that the developed Web pages can be viewed on monitors of different sizes.
- See if the Web pages can be downloaded in a reasonable amount of time over slower Internet connections.
- Stress-test the Web site, by simulating simultaneous use by hundreds or even thousands of people, to ensure that it still performs reasonably.
- Check for grammar and spelling errors.

These tests will produce bug reports, which your team will use to guide any necessary redesign, code changes, and simple fixes for the problems found during testing.

Implementation Deliverables

Completed Web Site • The completed Web site includes all of the HTML Web pages, all of the software, and all of the database tables required for the Web site to work.

Maintenance Document • The maintenance document describes in detail how to maintain the completed Web site. It explains which parts of the Web site will be periodically updated, how often they need to be updated, and who should update them. It should also describe the database tables, showing how they fit together.

Test Plan Document • The test plan spells out what steps will be taken to ensure that the Web site works as intended. At a minimum, it should include checking for performance, spelling, broken links, and the like. It should also explain how the site will be tested for usability, as well as how each of the features described in the specification document will be tested to ensure that they simply work.

Updates • Any documents that are out-of-date should be updated at this time (if they have not been already). These include the specification document, the design document, design guidelines, and Web page templates.

5.7 The Launch Phase

The Launch phase deals with the live deployment of the Web site. At this point, there is time to do only minor polishing on the Web site, such as checking for misspellings, grammatical errors, broken links, and broken images. All of the major checks should have been done in the previous phase, Implementation.

Some design teams choose to roll out the Web site in stages. Instead of developing the entire Web site at once and waiting until the very end to deploy it, selected parts of the Web site are created and deployed incrementally. Develop the most important functions and subsites first, and post them on a beta Web site for early adopters. Use the resulting feedback to drive the immediate design and development of the rest of the Web site. Handle staged deployment carefully because if the Web site is rolled out too early and shows a lack of content and polish, potential customers may avoid the site later.

Many design teams find a postmortem a useful exercise after launch, to assess what things went right, what did not, why, and how to avoid these problems in the future.

5.8 The Maintenance Phase

Maintenance is perhaps the most neglected aspect of Web site design. The objective of the Maintenance phase is to perform all of the activities needed to sustain a Web site. Beyond basic tasks such as updating the site with new content and promptly answering customer e-mail, Maintenance includes the following:

- Changing code and fixing bugs
- Collecting usability and satisfaction metrics
- Verifying that all links point to valid pages
- Checking that there are no spelling or grammatical errors
- Ensuring that pages in the Web site follow the design guidelines
- Periodically backing up the entire Web site (to a safe, distant computer)
- Updating the FREQUENTLY ASKED QUESTIONS (H7) pages
- Checking that your team is WRITING FOR SEARCH ENGINES (D6)
- Maintaining server logs that show where people come from, what search terms they use [see ORGANIZED SEARCH RESULTS (J3)], and what they're doing on the site

Maintenance includes assessing the Web site, collecting measurements on how customers use the Web site, analyzing and summarizing the metrics

you collect, and making the metrics available to the rest of the team and the company. Metrics are one of the most important parts of maintaining a Web site. They are the heartbeat of a Web site, measuring its overall health. Without them, you cannot tell which aspects are working and which are not.

The Maintenance Process

The Maintenance phase is the longest part of the design cycle. Consequently, it is also the most expensive—an often overlooked factor that must be considered when the budget is being developed. The most important decision in this phase, though, is determining when a revision to the Web site is necessary. A revision can be anything from a minor change to a complete overhaul. In any case, use all of the metrics, wisdom, and experience you have gained from developing and maintaining the old Web site to help ensure that the new Web site will succeed.

Small redesigns or additions to the existing site can go through an accelerated development process. In these cases you do not have to revisit such things as the business analysis document. However, pay attention to the specification document, the design document, and the design guidelines, with the goal of keeping the Web site's look and feel the same throughout. (Some design teams insist on making their work look different from the rest of the Web site. Unless there is an extremely compelling reason to the contrary, however, insist on maintaining a consistent look and feel for the entire Web site to make it easier for customers.)

On the other hand, complete overhauls mean going through the entire Web site development process again. Complete redesigns take place when major changes are needed, such as when customer expectations change, when customer behavior changes, when there are new technology considerations, when fresh content and functionality are necessary, or when the Web site starts looking outdated.

Get customer feedback before making the final switch to a new Web site. The best way is to run usability tests on your new site to verify that customers can complete the essential tasks. Another way is to provide a link on the homepage to let customers try out the new site and to ask them what they do and don't like. If you have a list of e-mail addresses, you can contact your customers to let them know that the Web site will be updated soon.

A third way of getting feedback is through virtual testing. In this case a few select customers will see the updated Web site instead of the regular Web site. You can compare what these customers do on the new Web site

to what customers do on the regular Web site to see if any interesting new strengths or weaknesses are evident with the new site. Amazon.com and Google use this method quite successfully. You can use online surveys for this purpose as well.

Maintenance Deliverables

Periodic Web Site Metrics • Most of your metrics will come from the business analysis document produced in the Discovery phase. They can include such things as the total number of hits, the conversion rate (the number of people who become paying customers), satisfaction metrics, and usability evaluations.

Bug Reports • Customer e-mails, Web site evaluations, and server log file analyses are good ways to find bugs. Rate each of these in terms of severity (such as “must fix,” “should fix,” and “could fix”), and estimate how long it will take to fix the problem.

Periodic Backups • Back up the entire Web site periodically as protection in case some files are accidentally erased, the Web servers are damaged, or hackers break into the site. Store the backups far away from the building that houses the Web server, in case of an environmental disaster like flood or fire.

5.9 Take-away Ideas

The process described in this chapter gives you a structure to make sure your Web site will meet the needs of your clients and the needs of their customers. The principles and techniques of customer-centered design and iterative prototyping are embedded in every stage. Many Web site design firms and internal design groups in companies have similar processes, though the stages and deliverables might have slightly different names. The names are not what's important here. The key is to make sure you have a customer-centered process that is documented, is reproducible, and can be improved by your organization over time.

